NATIONAL INSTITUTE OF TECHNOLGY PATNA

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

End Semester Examination July-Dec 2025

Max.Marks: 60 — ***CS54118 – Machine Learning***

Date:** December 2, 2025* — ***Time: 3 Hrs.

Instructions:

- Attempt all questions.
- Assume any suitable data, if necessary. (Any other Instruction need to provide by the concerned faculty)

	Questions	Marks	CO	BL
1	a) Explain the bias–variance tradeoff. b) How does the bias–variance tradeoff relate to underfitting and overfitting? c) What happens to bias and variance when model complexity increases? d) Why does regularization reduce variance? e) Describe how cross-validation helps in bias–variance tradeoff. f) For ridge regression, what happens to bias and variance when λ increases?	[12]	CO2 CO3	*Remember* *Apply*
2	What is computation graph. Draw the computation graph for the following function and find the gradients ∇_{x1} and ∇_{x2}. $f(x_1, x_2) = \left[\sin\left(\frac{x_1}{x_2}\right) + \frac{x_1}{x_2} - e^{x_2}\right] \times \left[\frac{x_1}{x_2} - e^{x_2}\right]$	[6]	CO4	*Apply*
3	Write and algorithm for Agglomerative hierarchical clustering. Cluster the following dataset using hierarchical clustering with group average links merging strategy and draw the dendrogram. X_1: 3, 5, 8, 9, 4, 2, 8, 8 X_2: 5, 8, 1, 3, 0, 6, 7, 4	[12]	CO3	*Implement* *Analyze* *Understand*
4	Write an algorithm for Linear discriminant analysis. Compute the Linear Discriminant projection for the following two-dimensional dataset. • Samples for class ω1: $X_1(x_1, x_2) = \{(4, 2), (2, 4), (2, 3), (3, 6), (4, 4)\}$ • Sample for class ω2: $X_2(x_1, x_2) = \{(9, 10), (6, 8), (9, 5), (8, 7), (10, 8)\}$	[10]	CO3	*Implement* *Analyze* *Understand*
5	Find the SVD of A, $U\Sigma V^T$, where $\mathbf{A} = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}$	[10]	CO3	*Implement* *Analyze* *Understand*
6	Consider a fully connected neural network with one hidden layer (using ReLU activation) and an output layer (using sigmoid activation) for binary classification. a) Derive the complete backpropagation equations for updating the weights between the hidden and output layers. Show how the gradient flows through the sigmoid function. b) Find the plane and compute the updated parameters of the model up to two epochs by using the learning rate **(α)**= 0.05 and the following initial random values (in your computation, roundup each value up to two digits): **θ_0=0.50, θ_1 = 0.75, θ_2= 0.25** for the following dataset. X_1: 1, 1, 2, 2, 3, 4 X_2: 1, 2, 1, 2, 2, 4 Y: +1, +1, +1, -1, -1, -1	[10]	CO4	*Implement* *Apply* *Analyze*

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NATIONAL INSTITUTE OF TECHNOLGY PATNA

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Mid Semester Examination July-Dec 2025

Max. Marks: 30 *CS54118 – Machine Learning*

Date: Sept 23, 2025 ***Time: 2 Hrs.***

Instructions:

1. Attempt all questions.
2. Assume any suitable data, if necessary. (Any other Instruction need to provide by the concerned faculty)

<table>
<tr><th></th><th>Questions</th><th>Marks</th><th>CO</th><th>BL</th></tr>
<tr><td>1</td><td>a) What is norm. Let $x = (1, -2, 3)^T$, then find the following norms.
a. $||x||_1$ b. $||x||_0$ c. $||x||_2$ d. $||x||_\infty$</td><td>[3]</td><td>CO1</td><td>Remember
Apply</td></tr>
<tr><td></td><td>b) Consider the following confusion matrix of a classifier., and compute the performance metrices like Accuracy, Recall, Precision, and F-score, TPR, FPR, FNR.
<table><tr><td></td><td colspan="4">Predicted condition</td></tr><tr><td>Actual condition</td><td></td><td>Class 1</td><td>Class 2</td><td>Class 3</td></tr><tr><td></td><td>Class 1</td><td>50</td><td>27</td><td>23</td></tr><tr><td></td><td>Class 2</td><td>35</td><td>40</td><td>25</td></tr><tr><td></td><td>Class 3</td><td>20</td><td>10</td><td>70</td></tr></table></td><td>[3]</td><td>CO2</td><td>Remember
Apply</td></tr>
<tr><td></td><td>c) Explain the Karush-Kunh-Trucker (KKT) conditions and Solve the following expression.
$min_{x,y}\ f(x, y) = x + y; s.t. x^2 + y^2 = 1$</td><td>[4]</td><td>CO2</td><td>Remember
Apply</td></tr>
<tr><td>2</td><td>Consider the dataset (as shown) with two independent attributes, X_1 and X_2 and a dependent variable, Y.
<table><tr><td>X_1</td><td>3</td><td>5</td><td>8</td><td>9</td><td>4</td><td>2</td><td>8</td><td>8</td></tr><tr><td>X_2</td><td>5</td><td>8</td><td>1</td><td>3</td><td>0</td><td>6</td><td>7</td><td>4</td></tr><tr><td>Y</td><td>8</td><td>5</td><td>8</td><td>5</td><td>7</td><td>9</td><td>6</td><td>4</td></tr></table>Apply the multiple linear regression model ($M_\theta(X)$) with Ridge regression and compute the updated parameters of the model up to two epochs by using the learning rate (α)= 0.05, regularization parameter (λ) = 2 and with the following initial random values (in your computation, roundup each value up to two digits): θ_0=0.50, θ_1 = 0.75, θ_2= 0.25.</td><td>[10]</td><td>CO2</td><td>Apply
Analyze
Understand</td></tr>
<tr><td>3</td><td>Derive the optimization problems for a Hard Margin Support Vector Machine. Construct the primal problem and solve this optimization problem using Lagrangian formulation in dual space. Express the final classifier in terms of optimal parameters. Consider the following 2-D linearly separable dataset.
<table><tr><td>X_1</td><td>1</td><td>1</td><td>2</td><td>2</td><td>3</td><td>4</td><td>5</td><td>5</td><td>5</td><td>6</td></tr><tr><td>X_2</td><td>1</td><td>2</td><td>1</td><td>2</td><td>2</td><td>4</td><td>3</td><td>4</td><td>5</td><td>5</td></tr><tr><td>Y</td><td>+1</td><td>+1</td><td>+1</td><td>+1</td><td>+1</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td></tr></table>a. Identify the support vectors.
b. Plot the data points, the decision boundary and the margin lines.</td><td>[10]</td><td>CO2</td><td>Implement
Apply
Analyze</td></tr>
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